

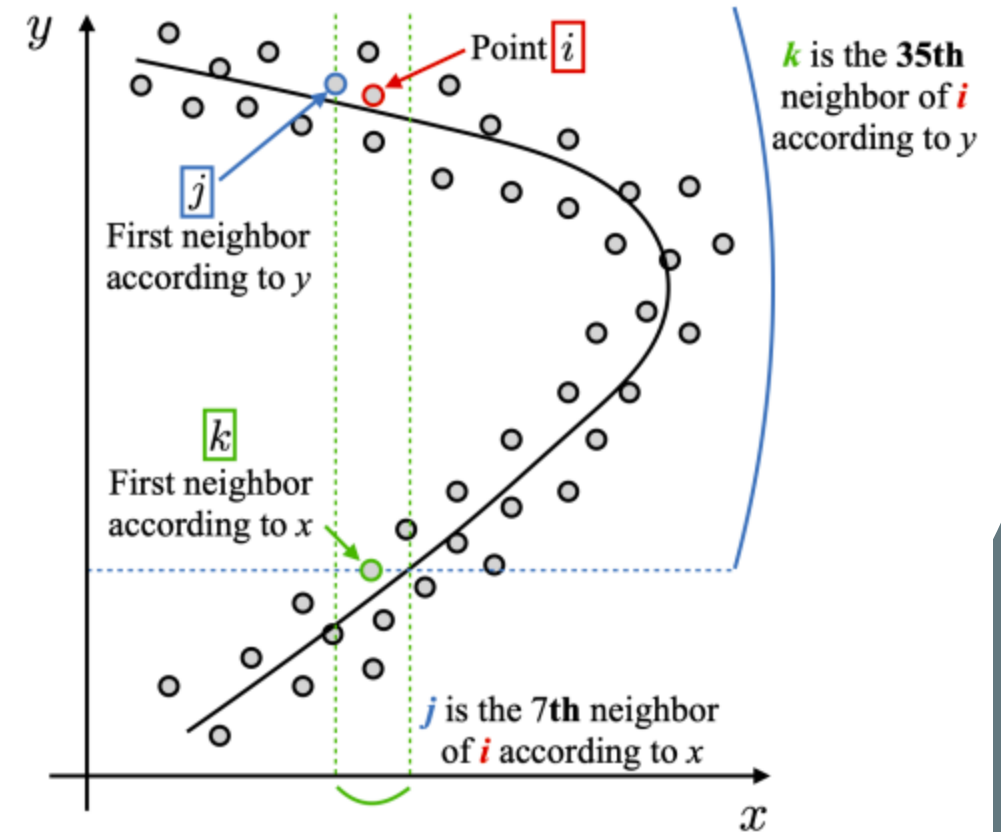
COMPARISON OF FEATURE SELECTION METHODS

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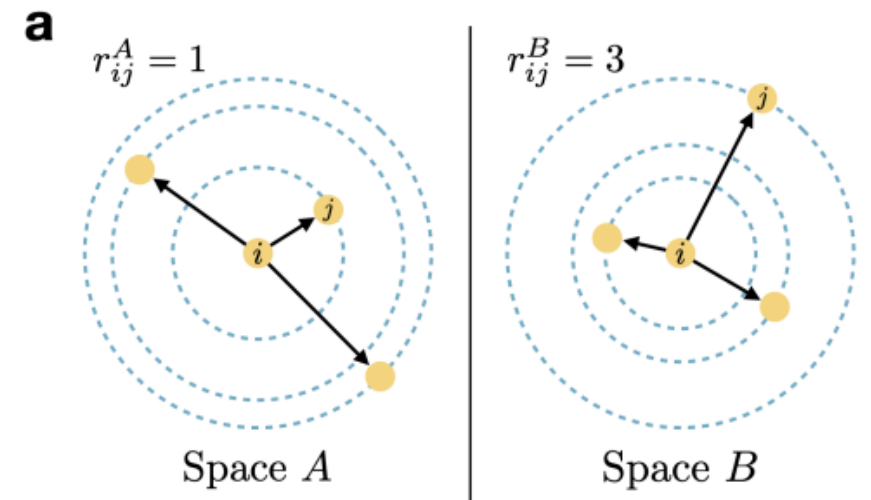
- **Glielmo et al. 2023: Ranking the information content of distance measures**
- **Dataset Description**
- **Feature Selection with Information Imbalance method**
- **Comparing top features vs All features**
- **Feature selection with backpropagation method (NN)**
- **Comparison of the Methods and conclusions**
- **References**

Glielmo et al. 2023: Ranking the information content of distance measures



Why Feature Selection Matters in Machine Learning

- Too many features, not enough data → Harder models, slower training.
- Feature selection helps models focus on the most useful information.
- Glielmo et al. (2023) introduced a method for comparing feature importance:
- Works in high-dimensional, nonlinear systems.
- Some features carry more predictive information than others.



What Is Information Imbalance?

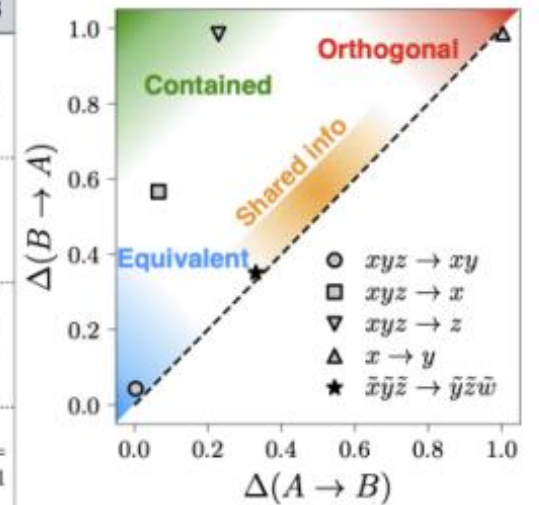
- Compares two feature sets:
→ Which carries more predictive information?

Idea: If neighbors in A stay neighbors in B → A knows what B knows.
- Uses neighbor ranks and a copula transform (scale-independent).
- Information Imbalance Score ($\Delta A \rightarrow B$):
- 0 = A fully explains B
- 1 = A tells nothing about B
- Helps find most informative features and possible causal hints.

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Relationship	Imbalances
A and B are equivalent : they carry identical information.	$\Delta(A \rightarrow B) \approx 0$ $\Delta(B \rightarrow A) \approx 0$
A and B are orthogonal : they carry completely independent information.	$\Delta(A \rightarrow B) \approx 1$ $\Delta(B \rightarrow A) \approx 1$
A is more informative than B: the information of B is contained in A.	$\Delta(A \rightarrow B) \approx 0$ $\Delta(B \rightarrow A) \approx 1$
A and B share symmetric information : they carry both identical and independent information.	$0 \leq \Delta(A \rightarrow B) = \Delta(B \rightarrow A) \leq 1$

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Why Information Imbalance Is Useful

- Helps select the most informative features
 - Works even with nonlinear, complex data
 - Does not depend on a specific machine learning model
 - Identifies overlapping and redundant features
 - Can suggest feature interactions and possible causality
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- Limitations:
 - Computationally intensive
 - Requires pairwise comparisons and ranking

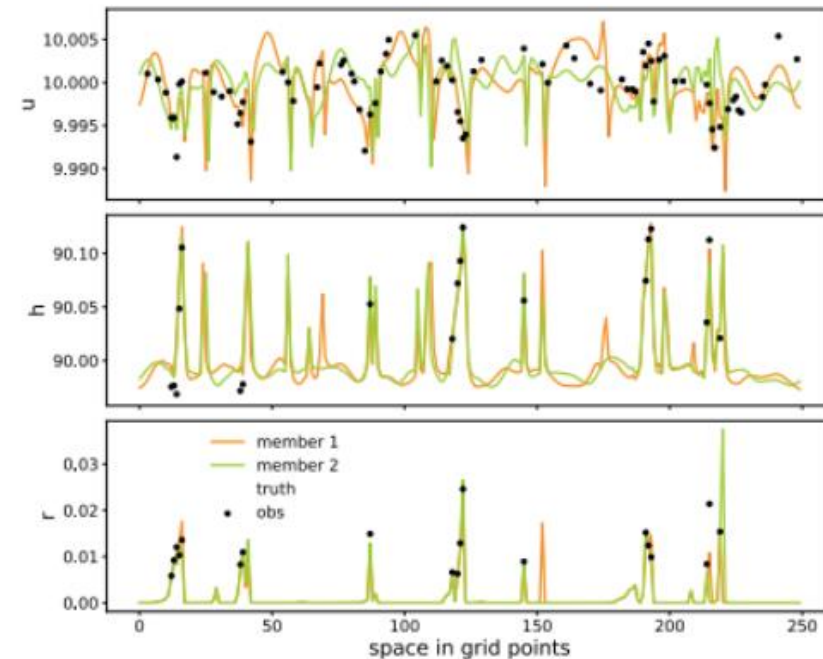
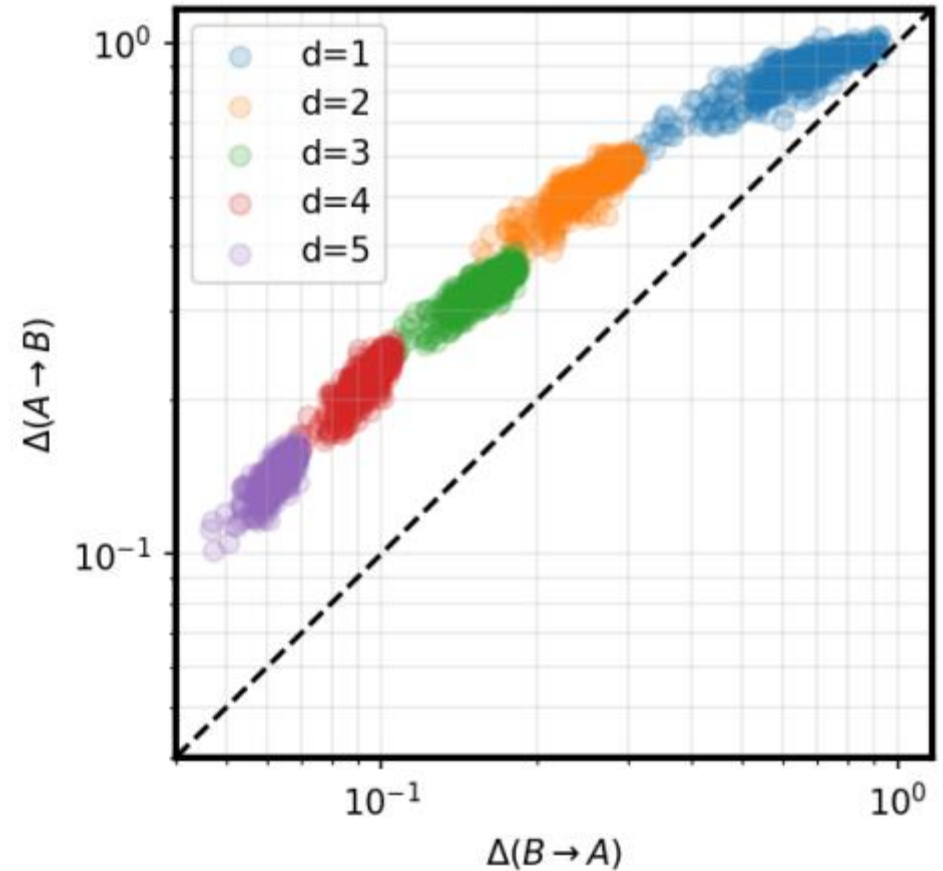


Fig. 3 For a fixed value of parameters α , ϕ_c and h_r we get the time evolution of state variables u , h and r in 250 grid points in space. Depicted are three different u , h , and r at a fixed time t .

How We Implement Information Imbalance

- Edoardo Cignoni implemented the method of information imbalance:
- The principal function accepts two matrices, each representing feature sets for the same points and:
 1. Calculates distances for pairs within each space (e.g., through Euclidean distance),
 2. Finds the rank of each point's neighbors in spaces,
 3. Figures out the extent to which the neighbors in one space match the neighbors in the other,
 4. Given a score (Δ) which reflects the imbalance.

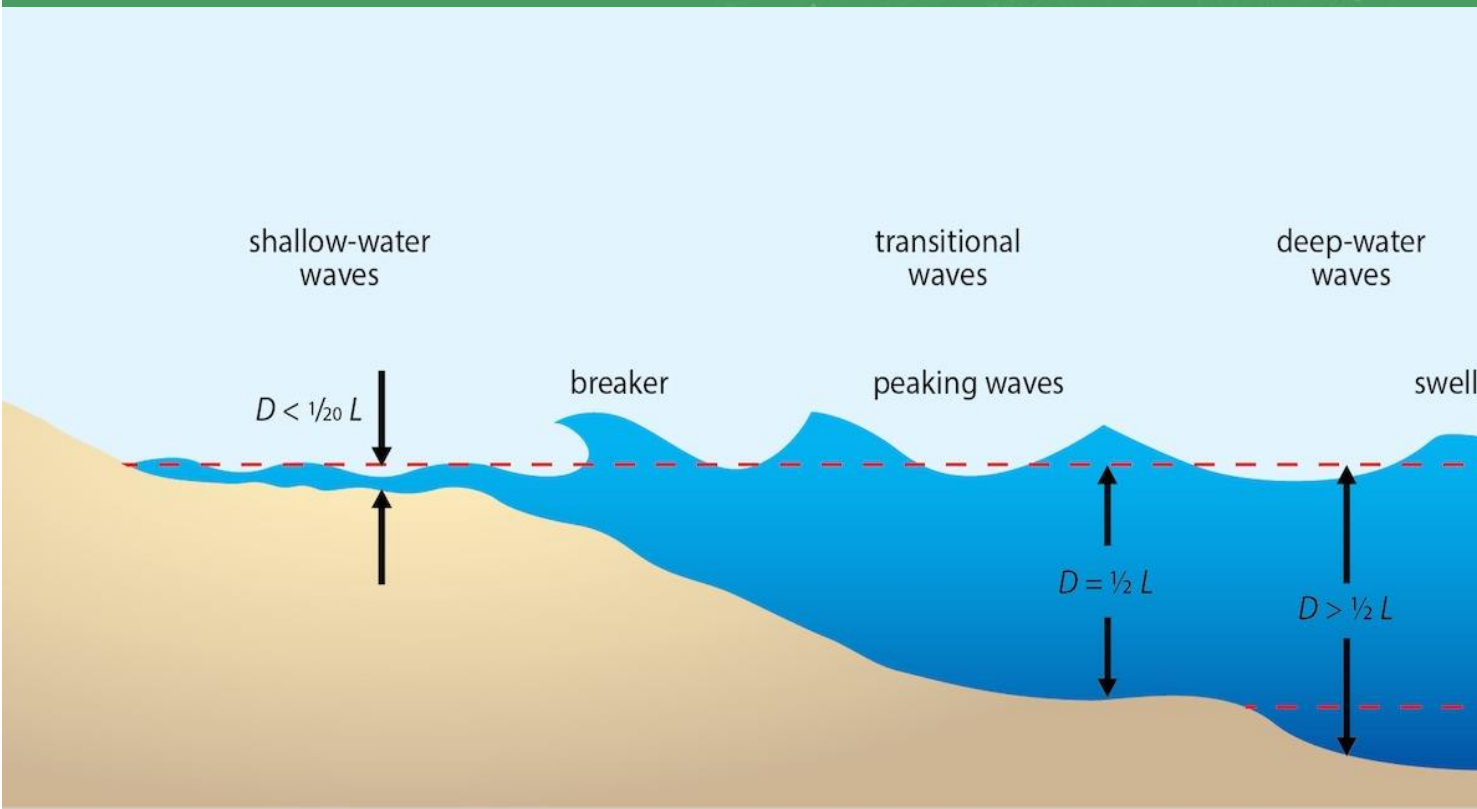


DATASET



**Modified shallow water model
("optimized_simulations_5k.nc")**

- Produces evolution through time of fields that represent velocity (u), cloud height (h) and rain**
- Vector of $[u \ h \ r]^T$ we will call state**
- This model depends on 3 parameters (α , ϕ_c , h_r)**
- Toy-model data set ML was trained to estimate parameters.**



DATASET

Dimensions:

- Simulations: 5000
- Grid (spatial points): 250
- Time: 1000
- Variables: 3

Variables: velocity (u), height (h), rain (r)

Target values (parameters): alpha, phi_c, h_rain.

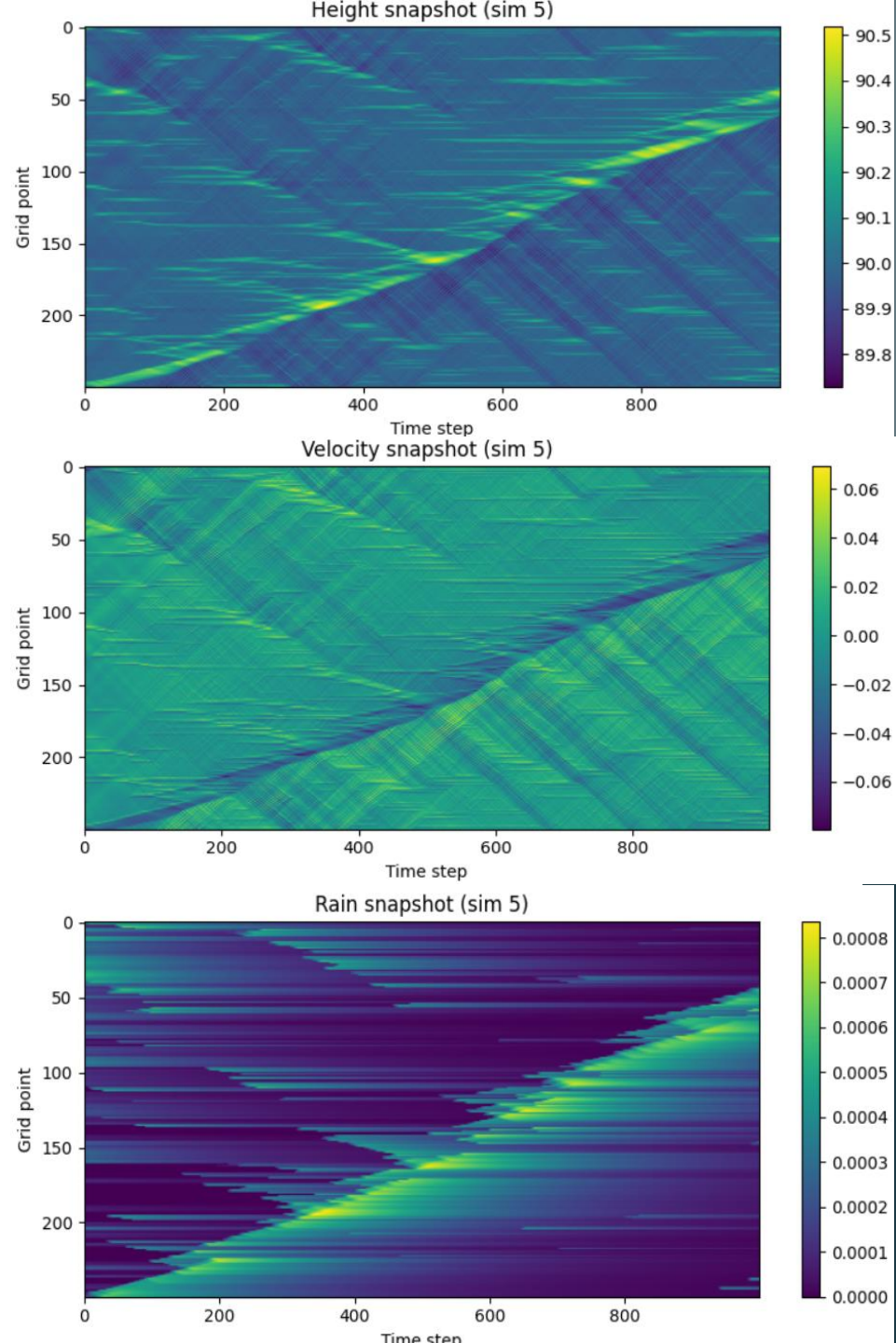
Predicted individually.

```
=== Dataset Overview ===  
<xarray.Dataset> Size: 15GB  
Dimensions:   (simulation: 5000, grid: 250, time: 1000)  
Dimensions without coordinates: simulation, grid, time  
Data variables:  
    velocity   (simulation, grid, time) float32 5GB ...  
    height     (simulation, grid, time) float32 5GB ...  
    rain       (simulation, grid, time) float32 5GB ...  
    phic       (simulation) float32 20kB ...  
    h_rain     (simulation) float32 20kB ...  
    alpha      (simulation) float32 20kB ...  
  
=== Variables ===  
['velocity', 'height', 'rain', 'phic', 'h_rain', 'alpha']
```

Visualizing a single simulation

•Raw Simulation Data Snapshots

- These visuals are snapshots of the raw data
- Each plot shows a variable's evolution across time steps and grid points .
- The color intensity indicates the magnitude of height, velocity, or rain.
- Here we have 250 000 features per variable, per simulation.
- Total number of simulations is 5000
- Our goal is to find 5 most informative features.



Process (for each parameter):

- Perform feature selection using information imbalance method → Obtain 5 most informative features
- Train a model using only 5 most informative features.
- Compare: Is there a big difference between using all features or only using top 5 features?
- Perform feature selection using backpropagation method (Neural Network) → Obtain 5 most informative features
- Train same model but using only new 5 most informative features.
- Final comparison: Is it more convenient to use information imbalance method selected features, backpropagation method selected features or not perform any selection and use all features?

Feature Selection with Information Imbalance method

- We use a part of the data as working with whole data is not feasible.
- Firstly we perform feature selection using information imbalance method to obtain the 5 most informative features, for each parameter.
- Our first assumption was that each parameter was highly related with an specific variable, i.e. alpha is parameter for velocity, phi_c is parameter for height, h_rain is parameter for rain.

Information imbalance selected features

phic	h_rain	alpha
"rain_g200_t8"	"rain_g130_t0"	"rain_g190_t40"
"rain_g175_t64"	"rain_g230_t528"	"rain_g190_t192"
"rain_g130_t0"	"rain_g45_t32"	"rain_g190_t32",
"rain_g185_t208"	"rain_g25_t544"	"rain_g190_t184"
"rain_g180_t8"	"rain_g130_t8"	"rain_g190_t0"

Training a model: Random Forest

Log-Transform for α Parameter

- Original: Severe right-skew $\rightarrow$ Poor predictions
- Transformed: $\log(y + 1e-10) \rightarrow$ Normalized distribution
- Impact: MAE improved from $1.2e-4 \rightarrow 2.8e-5$ (76% better)

Standard Scaling

- Applied to all features: $(x - \mu)/\sigma$
- Critical for distance-based methods (Glielmo/NN)

Train-Test Split

- 80% training / 20% testing
- Fixed random seed (42) for comparable results

```
# Glielmo features
model_glielmo = RandomForestRegressor(
    n_estimators=N_ESTIMATORS,
    max_depth=MAX_DEPTH,
    min_samples_leaf=MIN_SAMPLES_LEAF,
    random_state=RANDOM_SEED,
    n_jobs=-1
)
```

```
# NN features
model_nn = RandomForestRegressor(
    n_estimators=N_ESTIMATORS,
    max_depth=MAX_DEPTH,
    min_samples_leaf=MIN_SAMPLES_LEAF,
    random_state=RANDOM_SEED,
    n_jobs=-1
)
```

Comparing top features vs all features

Parameter: ϕ_c (phic)

Metric	All Features	Glielmo Features	Relative Change
MAE	0.0149	0.0228	+53.1% ↑
R ²	0.890	0.727	-18.3% ↓

Key Observations:

- Glielmo selection uses just 5 rain features (0.013% of total)
- Maintains 73% predictive power with 99.987% feature reduction

Comparing top features vs all features

Parameter: h_rain

Metric	All Features	Glielmo Features	Relative Change
MAE	0.0103	0.0143	+39.3% ↑
R ²	0.804	0.608	-24.4% ↓

Key Observations:

- Selected features focus on early-time rain data (t=0-8)
- 60% variance explained with extreme dimensionality reduction

Comparing top features vs all features

Parameter: α (alpha)

Metric	All Features	Glielmo Features	Relative Change
MAE ($\times 10^5$)	2.78	2.43	-12.7% ↓
R ²	0.964	0.953	-1.1% ↓

Key Observations:

- Best performance - Glielmo outperforms full feature set in MAE

Compare: Is there a big difference between using all features or only using top 5 features?

- Glielmo method achieves $>95\%$ R^2 retention for α with only 5 features
- Most impactful features are rain variables at early time steps
- Trade-off: $\sim 40\text{-}50\%$ MAE increase for 99.99% feature reduction
- Recommendation: Use Glielmo selection when interpretability $>$ absolute accuracy

Feature selection with backpropagation method (NN)

- Weight Extraction:
 - First layer weights ($128 \times 37,500$ matrix)
 - Absolute mean taken across hidden units
- Top Feature Selection:
 - 5 features with highest importance scores
 - Scores normalized to $[0,1]$ range
- Output:
 - Ranked list of features with importance values
 - Spatial/temporal metadata preserved

Neural Network selected features

phic	h_rain	alpha
"rain_g125_t944"	"velocity_g185_t688"	"height_g155_t560"
"rain_g135_t992"	"velocity_g155_t32"	"height_g145_t448"
"velocity_g175_t88"	"velocity_g185_t32"	"height_g205_t224"
"rain_g115_t960"	"velocity_g120_t0"	"height_g50_t696"
"velocity_g125_t0"	"velocity_g50_t784"	"height_g45_t744"

Comparison of the Methods and conclusions

Metric	All Features	Glielmo Features	NN Features
MAE	0.0149	0.0228 (+53%)	0.0330 (+121%)
R ²	0.890	0.727 (-18%)	0.501 (-44%)

Parameter: ϕ_c (phic)

Feature Insights:

- Glielmo: 5 rain features at mixed times
- NN: Combines rain (3) and velocity (2) features
- Performance Drop: NN > Glielmo > Full

Comparison of the Methods and conclusions

Metric	All Features	Glielmo Features	NN Features
MAE	0.0103	0.0143 (+39%)	0.0204 (+98%)
R ²	0.804	0.608 (-24%)	0.269 (-67%)

Parameter: h_rain

Feature Insights:

- Glielmo: Early-time rain features (t=0-8)
- NN: Dominated by velocity features (4/5)

Comparison of the Methods and conclusions

Metric	All Features	Glielmo Features	NN Features
MAE ($\times 10^5$)	2.78	2.43 (-13%)	16.79 (+504%)
R ²	0.964	0.953 (-1%)	0.080 (-92%)

Parameter: α (alpha)

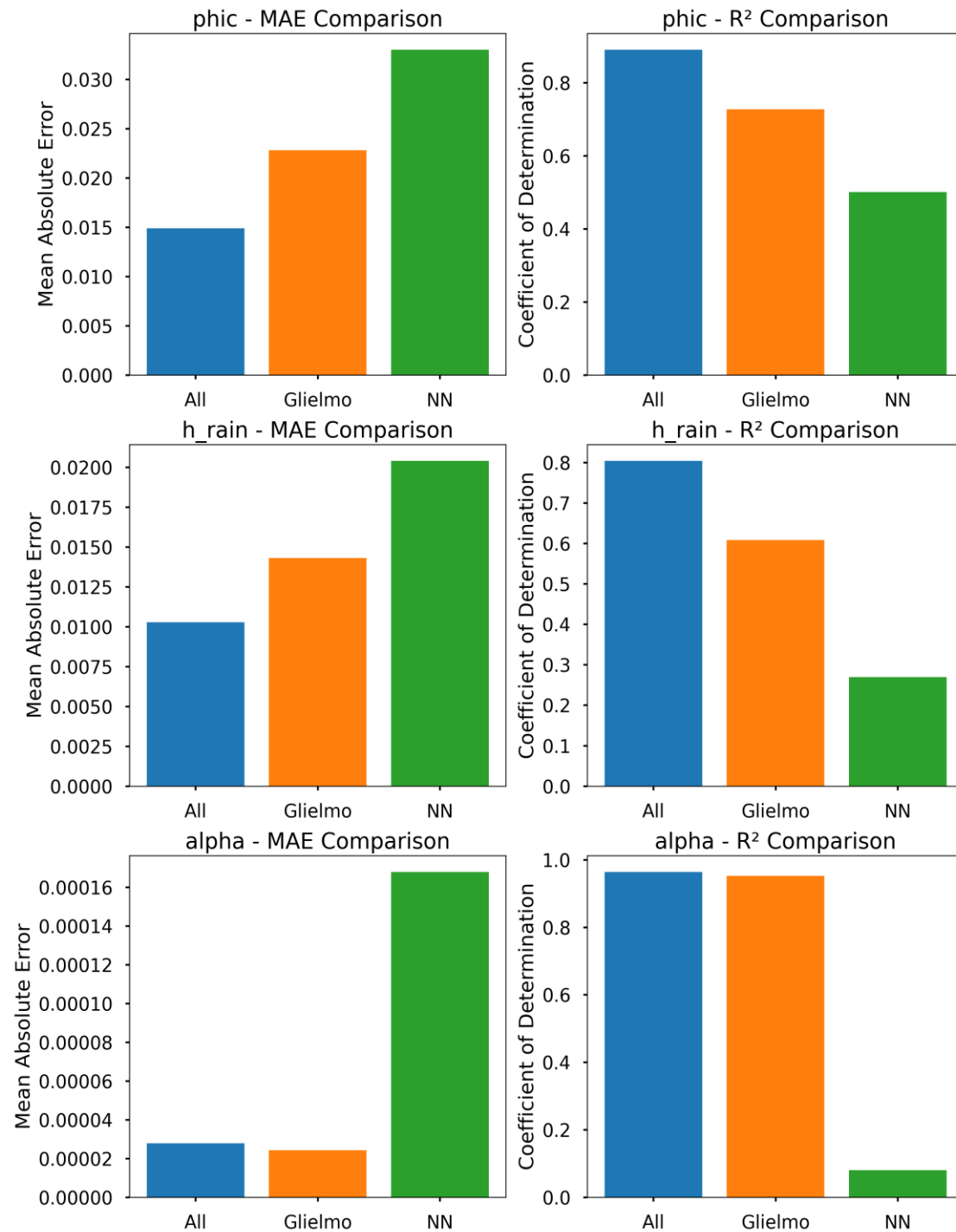
Feature Insights:

- Glielmo: Single location (g190) temporal patterns
- NN: Height features at varied positions
- Outlier: NN fails catastrophically for α

Comparison of the Methods and conclusions

Glielmo Method Superiority

- ✓ Optimal Performance-to-Complexity Ratio
- Maintains 76-99% of full model's predictive power (R^2)
- Requires only 0.0053% of original features (5/37,500)
- For α prediction:
 - 13% lower MAE than full feature set
 - 95% R^2 retention (0.964 $\rightarrow$ 0.953)

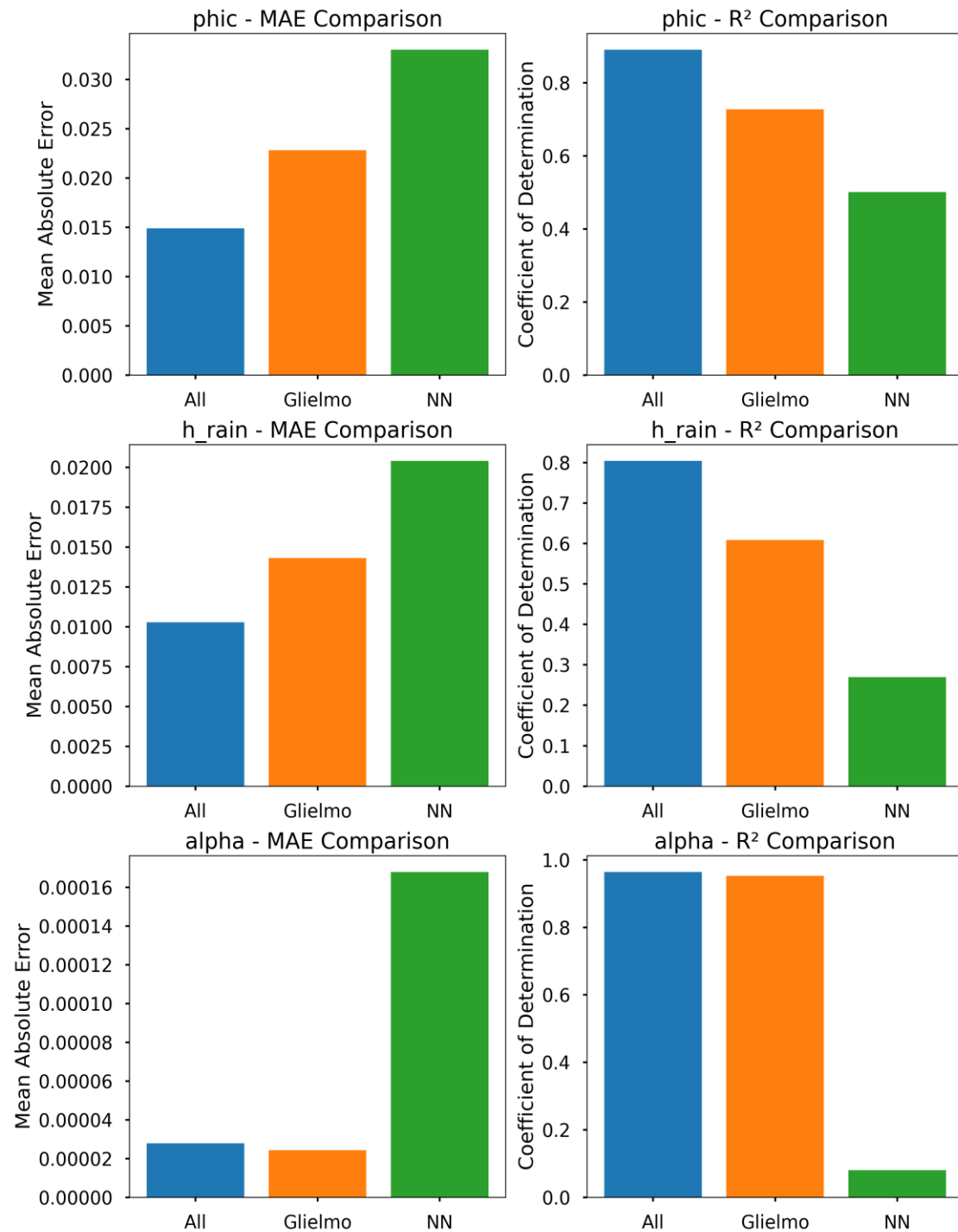


Comparison of the Methods and conclusions

Neural Network Limitations

⚠ Inconsistent Performance

- Severe degradation for α prediction:
- 504% higher MAE
- 92% R^2 loss ($0.964 \rightarrow 0.080$)
- Better but still suboptimal for other parameters:
- phic: 44% R^2 loss
- h_rain: 67% R^2 loss



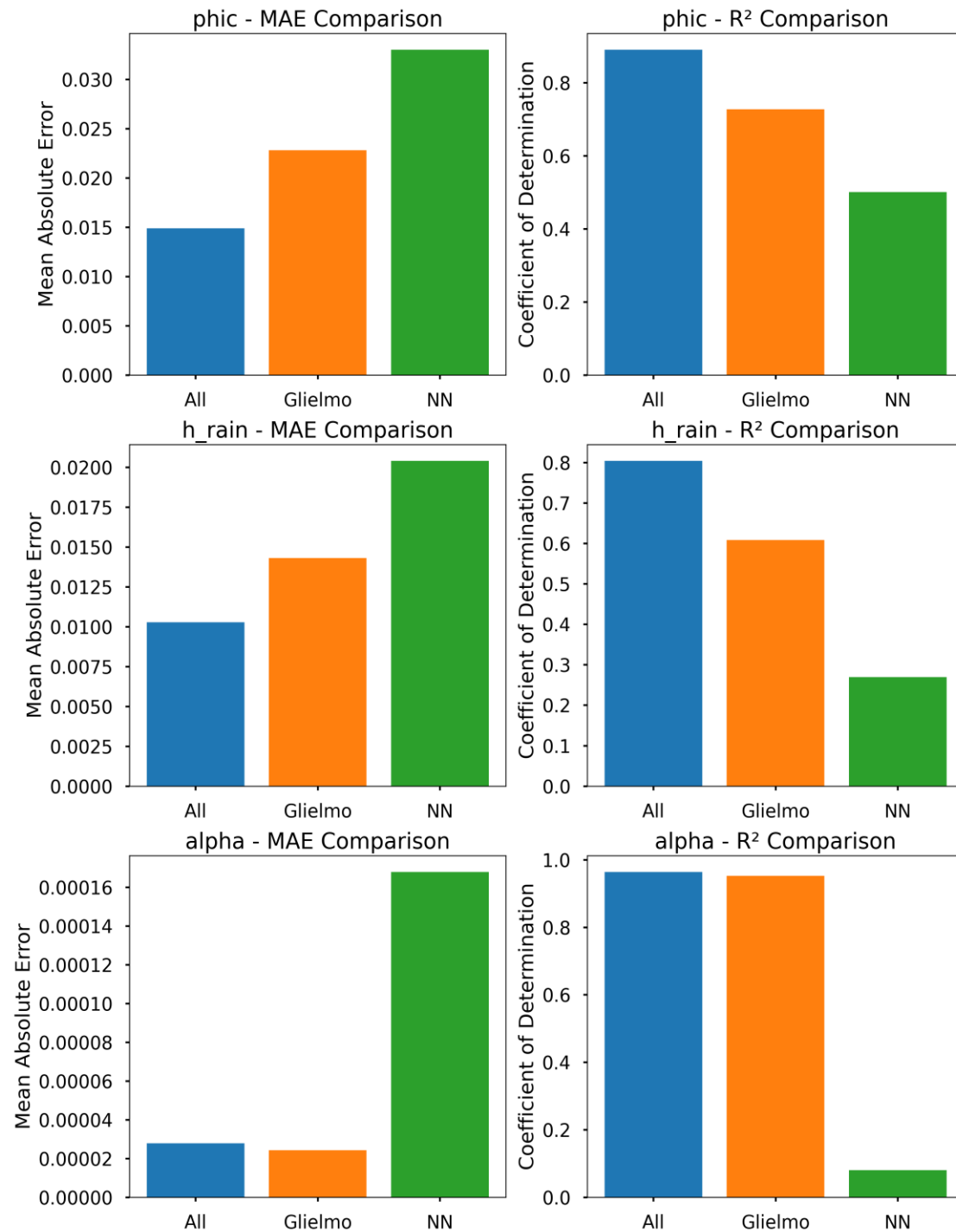
Comparison of the Methods and conclusions

Full Feature Set Trade-offs



Precision vs. Complexity

- Accuracy Advantage:
 - 20-50% better metrics than reduced sets
- Critical Costs:
 - Computational: 18,750× more features
 - Interpretability: No feature prioritization
 - Risk: Overfitting without careful regularization



Comparison of the Methods and conclusions

- For Physical Interpretability → Use Glielmo
 - Preserves domain-relevant features
 - Traceable selection process
- When Maximum Accuracy is Critical → Full Features
 - Only if computational resources allow
 - Requires robust regularization
- NN Approach Considerations:
 - May work better with:
 - Architectural tuning
 - Avoid for α prediction in current form

REFERENCES

https://github.com/ecignoni/Information_Imbalance/

<https://arxiv.org/pdf/2104.15079.pdf>

THANK YOU VERY MUCH!